

RAJKUMAR PUJARI

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PhD in NLP | LLM Post-Training, Reinforcement Learning & Evaluation

RESEARCH INTERESTS

LLM post-training and reinforcement learning; reward modeling and evaluation; reasoning and representation learning; experimental design for subjective and underspecified domains.

EDUCATION

Purdue University

Ph.D. in Computer Science (Natural Language Processing)

West Lafayette, IN

Aug. 2017 – Aug 2025

- **Thesis Topics:** Contextualized Discourse Representations, ML for Social and Cultural Understanding.
- **Advisor:** Prof. Dan Goldwasser | **GPA:** 3.8/4.0

Indian Institute of Technology (IIT) Kharagpur

B.Tech (Hons.) in Computer Science and Engineering | GPA: 9.1/10

West Bengal, India

May 2014

TECHNICAL SKILLS

RL & Post-Training: PPO, GRPO, DPO, Slime, Vime, rubric judges, reward modeling, human-in-the-loop, synthetic data collection & filtering

Training & Inference: PyTorch, Megatron-LM, vLLM, CUDA, SageMaker, Slurm, Kubernetes

NLP & Research: discourse modeling, pragmatics, political & social-context NLP, stereotype/bias detection, LLM-as-judge evaluation

EXPERIENCE

Amazon AGI

Applied Scientist II (Post-training, Nova Models)

Sunnyvale, CA

Aug. 2025 – Present

- **Rubric-Based LLM Judges:** Designed and trained a family of calibrated LLM judges that generate, follow, or complete a (partial) rubric via a two-stage curriculum: SFT on rubric-enriched data → GRPO with rewards for preference alignment, score-confidence calibration, and positional consistency. Built the supporting synthetic-data generation and filtering pipeline. The best judge outperforms DeepSeek-V3 and is comparable to Claude Sonnet-4.5 across 10 preference benchmarks.
- **Judge-Guided Model Optimization & Deployment:** Used the trained judges as a selection signal to iteratively evaluate and optimize Nova 2 recipes and checkpoints, improving chat quality by **+90 LMArena Elo** while replacing a prior GPT-4-based evaluation dependency. Launched two judge variants on SageMaker for external customers.
- **Model Spec Judge Evaluation:** Built and owned an automated pipeline that benchmarks candidate judges against ≈ 100 behavioral traits using win-rate-based Elo correlation (Ridge regression over per-trait win rates) and synthetic gold-vs-corrupted trait discrimination. The pipeline drove judge selection for both offline evaluation and rubric-based RL.
- **Nova 2 Recipe & Distillation:** Quantified the Elo contribution of individual SFT levers through structural ablations, including Arena-prompt upsampling, agentic/SWE data, hard/on-policy distillation, and Arena-specific RL.
- **Reasoning Optimization (in progress):** Designed the experimental and human-in-the-loop annotation pipeline for a learned semantic-step segmentation model on GPT-OSS-20B, targeting process-reward RL, reasoning-quality measurement, and reasoning compression.
- **LLM Creativity:** Collaborated with academic researchers on AGC-Bench (arXiv:2607.01152), an agentic pipeline that operationalizes the creativity literature into HELM-standardized benchmarks. Analysis across 83 LLMs finds that a single latent creativity factor explains **81.5%** of performance variance.
- **LMArena Serving Infrastructure:** Own an end-to-end SageMaker pipeline for serving Nova Micro checkpoints to the public LMArena leaderboard. Validated full 128k-context serving at concurrency 8 on a single ml.g6.48xlarge, enabling deployment of ≈ 10 experimental checkpoints/week.

Microsoft Research

Research Intern (JEM Team)

Redmond, WA

May 2021 – Aug. 2021

- **RL for Safety:** Developed a Reinforcement Learning guided Multitask Learning framework for low-resource Stereotype Detection.
- **Data Efficiency:** Leveraged neighboring toxicity datasets (Hate-Speech, Misogyny) to systematically improve zero-shot and few-shot performance on stereotype detection.
- **Outcome:** Published work at **ACL 2022** (Main Conference).

Amazon Alexa AI

Applied Scientist Intern

Sunnyvale, CA

May 2019 – Aug. 2019

- Designed a slot-filling based generative neural architecture for Conversational Question Answering in the sports domain.
- Built a ParlAI-style data collection tool and curated a novel conversational QA dataset via MTurk.

Indian Institute of Technology (IIT) Bombay – CFILT Lab

Research Assistant

Mumbai, India

Jan. 2016 – Apr. 2017

- Worked on semantic question matching for customer service use cases at the Center for Indian Language Technology (CFILT) lab, under Prof. Pushpak Bhattacharyya and Dr. Asif Ekbal.
- Work resulted in publication at **COLING 2018**.

- Developed predictive strategies (alphas) for algorithmic trading, optimizing for high Sharpe ratios and low turnover.
- Promoted to Senior Researcher within 1 year for exceptional performance.

PUBLICATIONS

Google Scholar

Peer-Reviewed Publications

- [1] S. Roy*, **R. Pujari***, T. Kumarage, C. Peris, R. Gupta, A. Rumshisky, P. Natarajan, V. Saligrama. *PREMISE: Policy Rubrics as Measurement Specifications for LLM Judges*. **EMNLP 2026 main**. *Equal contribution.
- [2] N. S. Nakshatri, E. Caplan, **R. Pujari**, D. Goldwasser. *TAIGR: Towards Modeling Influencer Content on Social Media via Structured, Pragmatic Inference*. **ACL 2026 main** (Oral).
- [3] **R. Pujari**, D. Goldwasser. *LLM-Human Pipeline for Cultural Context Grounding of Conversations*. **NAACL 2025 main** (Oral).
- [4] **R. Pujari**, C. Wu, D. Goldwasser. *"We Demand Justice!": Towards Grounding Political Text in Social Context*. **EMNLP 2024 main** (Oral).
- [5] **R. Pujari**, E. Oveson, P. Kulkarni, E. Nouri. *Reinforcement Guided Multi-Task Learning Framework for Low-Resource Stereotype Detection*. **ACL 2022 main** (Poster).
- [6] **R. Pujari**, D. Goldwasser. *Understanding Politics via Contextualized Discourse Processing*. **EMNLP 2021 main** (Oral).
- [7] **R. Pujari**, D. Goldwasser. *Using Natural Language Relations between Answer Choices for Machine Comprehension*. **NAACL-HLT 2019** (Oral).
- [8] D. Gupta, **R. Pujari**, A. Ekbal, P. Bhattacharyya, A. Maitra, T. Jain, S. Sengupta. *Can Taxonomy Help? Improving Semantic Question Matching using Question Taxonomy*. **COLING 2018** (Poster).
- [9] A. Maitra, S. Sengupta, A. Mukhopadhyay, D. Gupta, **R. Pujari**, et al. *Semantic Question Matching in Data Constrained Environment*. **Text, Speech and Dialogue (TSD) 2018**.
- [10] **R. Pujari**, S. Desai, N. Ganguly, P. Goyal. *A Novel Two-Stage Framework for Extracting Opinionated Sentences from News Articles*. **TextGraphs-9 Workshop @ EMNLP 2014**.

Preprints & Papers Under Review

- [11] R. Beaty, V. Deshpande, C. K. Y. Lai, A. Attuch, N. Shivagunde, S. Roy, **R. Pujari**, P. V. DiStefano, S. Muckatira, C. E. Stevenson, M. Gronas, A. Rumshisky. *AGC-Bench: Measuring Artificial General Creativity*. Under review. [arXiv:2607.01152](https://arxiv.org/abs/2607.01152)

ACADEMIC SERVICE

Program Committee / Reviewer: ACL (since 2019), NAACL (since 2019), EMNLP (since 2019), AAAI, IJCAI, COLM, COLING, LREC.
ACL Rolling Review (ARR): Reviewer, 20+ review cycles (2022 – 2026).

AWARDS & HONORS

Outstanding Reviewer, ACL-IJCNLP 2021.